

(Applications Development and Emerging Technologies)

TECHNICAL-SUMMATIVE ASSESSMENT

2

PHP ARRAYS AND PRE-DEFINED FUNCTIONS

Student Name / Group Name:	Franzese Denn B. Selosa	
Members (if Group):	Name	Role
Section:	TC23	
Professor:	Mr. John Benedict Legaspi	

I. PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.

II. COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Understand and apply best practices and standards in the development of website.

III. INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- To know the different approach of using arrays in PHP.
- To use PHP lazy function foreach to iterate through array elements.
- To implement one dimensional and multi-dimensional in the program.
- To know use variables that is globally declared and locally declared.
- To know how to include separate PHP code in the main page for code enhancement.
- To be familiar with the use of common predefined function such as define, include, and require.
- To use different available mathematical function for manipulating numbers.

IV. BACKGROUND INFORMATION

PHP Arrays and User Defined Functions

Array: one dimensional

Example: Code 1

```
<?php
$fruit[0] = "orange";
$fruit[1] = "apple";
$fruit[2] = "grapes";
$fruit[3] = "banana";
echo "<pre>";
print_r($fruit);
echo "</pre>";
?>
```

Example: Code 2

```
<?php
$fruit[] = "orange";
$fruit[] = "apple";
$fruit[] = "grapes";
$fruit[] = "banana";
echo "<pre>";
print_r($fruit);
echo "</pre>";
?>
```

Example: Code 3

```
<?php
$fruit = array("orange", "apple", "grapes", "banana");
echo "<pre>";
print_r($fruit);
echo "</pre>";
?>
```

Output:

```
Array
(
    [0] => orange
    [1] => apple
    [2] => grapes
    [3] => banana
)
```

PHP Predefined Functions

Using Constant

define() functions

used to declare constants. A constant can only be assigned a scalar value, like a string or a number. A constant's value cannot be changed.

Syntax:

```
define('NAME', 'value');
```

Example:

```
<?php
define('MAX_VALUE', 10);
for($i=5; $i<MAX_VALUE; $i++){
    echo $i." ";
}
//MAX_VALUE = 20; is invalids
?>
```

Output:

```
5 6 7 8 9
```

V. GRADING SYSTEM / RUBRIC (please see separate sheet)

VI. LABORATORY ACTIVITY

1. **Using ARRAYS**, create a fruit directory that will show the image, name, description and facts about the fruit (create at least 10 fruits sort in alphabetical order). Implement a HTML and CSS code for the structure and design of the webpage. Please see the sample output below.

My Fruits			
Image	Name	Description	Facts
	Banana	Color Yellow	Bananas are a healthful addition to a balanced diet, as they provide a range of vital nutrients and are a good source of fiber.

2. **USING USER DEFINED FUNCTION.** Create a formula for each volume of shapes (Cube, Right Rectangular Prism, Prism or Cylinder, Pyramid or Cone and Sphere) then display the result. Please see the sample below:

Volume of Shapes		
Values	Formula	Answer
$s = 5$	$V=s^3$	125

3. Convert your Student Resume into a web based form, used include() and require() functions to connect all pages

Include and create the following pages:

- Career Objective page
- Personal information page
- Educational Attainment page
- Skills page
- Affiliation page
- Work Experience Page

Please see the sample below:

	Personal information
	• Career Objective
	• Educational Attainment page
	• Skills page
	• Affiliation page
	• Work Experience Page

Code 1:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Technical 2 Act 1</title>
</head>
<body>
  <h1>Fruit Directory</h1>

  <table border="1" cellspacing="0" cellpadding="10" width="100%">
    <tr>
      <th colspan="4" style="text-align:center;">Fruits ni Denn</th>
    </tr>
    <tr>
      <th>Image</th>
```

```

        <th>Name</th>
        <th>Description</th>
        <th>Facts</th>
    </tr>

    <?php
    $fruits = [
    [
        "name" => "Apple",
        "image" => "images/Apple.jpg",
        "description" => "Apples are sweet, crisp fruits commonly red, green,
or yellow.",
        "fact" => "Apples float in water because they are 25% air."
    ],
    [
        "name" => "Banana",
        "image" => "images/Banana.jpg",
        "description" => "Bananas are soft, sweet fruits wrapped in a yellow
peel.",
        "fact" => "Bananas are technically berries, while strawberries are
not."
    ],
    [
        "name" => "Cherry",
        "image" => "images/Cherry.jpg",
        "description" => "Cherries are small, round fruits that can be sweet
or tart.",
        "fact" => "Japan's cherry blossoms (sakura) are world famous."
    ],
    [
        "name" => "Durian",
        "image" => "images/Durian.jpg",
        "description" => "Durian is a large, spiky fruit known for its strong
smell.",
        "fact" => "It's banned in many hotels and airports in Southeast
Asia."
    ],
    [
        "name" => "Grape",
        "image" => "images/Grape.jpg",
        "description" => "Grapes are juicy fruits that come in clusters and
various colors.",

```

```

        "fact" => "Grapes are used to make wine, juice, raisins, and
vinegar."
    ],
    [
        "name" => "Kiwi",
        "image" => "images/Kiwi.jpg",
        "description" => "Kiwi is a fuzzy brown fruit with bright green flesh
and tiny black seeds.",
        "fact" => "Kiwis are rich in Vitamin C—more than oranges."
    ],
    [
        "name" => "Mango",
        "image" => "images/Mango.jpg",
        "description" => "Mangoes are sweet, tropical fruits with golden
flesh and a single seed.",
        "fact" => "India is the largest producer of mangoes in the world."
    ],
    [
        "name" => "Orange",
        "image" => "images/Orange.jpg",
        "description" => "Oranges are citrus fruits known for their bright
skin and juicy flesh.",
        "fact" => "Oranges are the most cultivated fruit tree in the world."
    ],
    [
        "name" => "Pineapple",
        "image" => "images/Pineapple.jpg",
        "description" => "Pineapples are tropical fruits with rough spiky
skin and sweet flesh.",
        "fact" => "It takes almost 2 years for a pineapple to grow"
    ],
    [
        "name" => "Strawberry",
        "image" => "images/Strawberry.jpg",
        "description" => "Strawberries are red, juicy fruits with seeds on
the outside.",
        "fact" => "Strawberries are the only fruit with seeds on the
outside."
    ],
    [
        "name" => "Lemon",
        "image" => "images/Lemon.jpg",

```

```

        "description" => "A tart citrus fruit widely used for flavoring and
cleaning.",
        "fact" => "Lemons are rich in vitamin C and aid digestion."
    ],
    [
        "name" => "Papaya",
        "image" => "images/Papaya.jpg",
        "description" => "A tropical fruit with buttery texture and sweet
flavor.",
        "fact" => "Papayas aid digestion with an enzyme called papain."
    ],
    [
        "name" => "Watermelon",
        "image" => "images/Watermelon.jpg",
        "description" => "Large, juicy fruit that's perfect for hydration.",
        "fact" => "Watermelon is 92% water and great for hydration."
    ],
    [
        "name" => "Dragon Fruit",
        "image" => "images/Dragonfruit.jpg",
        "description" => "Dragon Fruit has vibrant pink skin and speckled
flesh, known for its mildly sweet taste.",
        "fact" => "Also called pitaya, it's a cactus fruit rich in fiber and
antioxidants."
    ],
    [
        "name" => "Rambutan",
        "image" => "images/Rambutan.jpg",
        "description" => "Rambutan is a hairy red fruit with juicy,
translucent flesh inside.",
        "fact" => "Despite its exotic look, rambutan is closely related to
lychee and is rich in vitamin C."
    ],
    [
        "name" => "Atis",
        "image" => "images/Atis.jpg",
        "description" => "Atis is a green, scaly fruit with creamy, sweet
white pulp and many seeds.",
        "fact" => "Also called sugar apple, Atis is popular in tropical
countries and rich in dietary fiber."
    ],
    [

```



```

        "name" => "Jackfruit",
        "image" => "images/Jackfruit.jpg",
        "description" => "Jackfruit is a large tropical fruit with spiky skin
and sweet, yellow flesh.",
        "fact" => "Jackfruit is the largest tree-borne fruit and is often
used in both desserts and savory dishes in Filipino cuisine."
    ],
    [
        "name" => "Santol",
        "image" => "images/Santol.jpg",
        "description" => "Santol has a thick rind and soft, white pulp that
can be sweet or sour.",
        "fact" => "In the Philippines, santol is often enjoyed with salt or
bagoong as a sour snack."
    ],
    [
        "name" => "Guyabano",
        "image" => "images/Guyabano.jpg",
        "description" => "Guyabano has spiky green skin and soft white flesh
with a sweet and tangy flavor.",
        "fact" => "Guyabano is believed to have medicinal properties and is
often used in juices and smoothies."
    ],
    [
        "name" => "Coconut",
        "image" => "images/Coconut.jpg",
        "description" => "Coconut is a tropical fruit with a hard shell,
white meat, and refreshing water inside.",
        "fact" => "Every part of the coconut tree is used in Filipino
culture—from food and drink to building materials and crafts."
    ]
];

usort($fruits, fn($a, $b) => strcmp($a['name'], $b['name']));

foreach ($fruits as $fruit) {
    echo "<tr>";
    echo "<td><img src='{${fruit['image']}}' alt='{${fruit['name']}}'
class='fruit-img'></td>";
    echo "<td class='name'>${fruit['name']}</td>";
    echo "<td class='description'>${fruit['description']}</td>";
    echo "<td class='fact'>${fruit['fact']}</td>";
}

```

```

        echo "</tr>";
    }
    ?>
</table>
</body>
</html>

<style>
    body {
        font-family: Arial, sans-serif;
        background: #eef6ff;
        margin: 0;
        padding: 20px;
    }

    h1 {
        text-align: center;
        color: #2c3e50;
    }

    table {
        border-collapse: collapse;
        width: 100%;
        background-color: #fff;
    }

    th, td {
        border: 1px solid #000;
        vertical-align: top;
    }

    th {
        background-color: #f2f2f2;
    }

    td.name, td.description, td.fact {
        text-align: center;
        vertical-align: middle;
        padding: 10px;
    }

    .fruit-img {

```

```

width: 100px;
height: 100px;
object-fit: cover;
display: block;
margin: 0 auto;
}
</style>

```

Output:

Fruit Directory			
Fruits ni Denn			
Image	Name	Description	Facts
	Apple	Apples are sweet, crisp fruits commonly red, green, or yellow.	Apples float in water because they are 25% air.
	Atis	Atis is a green, scaly fruit with creamy, sweet white pulp and many seeds.	Also called sugar apple, Atis is popular in tropical countries and rich in dietary fiber.
	Banana	Bananas are soft, sweet fruits wrapped in a yellow peel.	Bananas are technically berries, while strawberries are not.
	Cherry	Cherries are small, round fruits that can be sweet or tart.	Japan's cherry blossoms (sakura) are world famous.
	Coconut	Coconut is a tropical fruit with a hard shell, white meat, and refreshing water inside.	Every part of the coconut tree is used in Filipino culture—from food and drink to building materials and crafts.
	Dragon Fruit	Dragon Fruit has vibrant pink skin and speckled flesh, known for its mildly sweet taste.	Also called pitaya, it's a cactus fruit rich in fiber and antioxidants.

Code 2:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Technical 2 Act 2</title>
</head>
<body>

  <h1>Volume of Shapes</h1>

  <table>
    <tr>
      <th>Shape</th>
      <th>Formula</th>
      <th>Values</th>
      <th>Volume</th>
    </tr>

    <?php
    function volumeCube($side) {
      return pow($side, 3);
    }

    function volumeRectangularPrism($length, $width, $height) {
      return $length * $width * $height;
    }

    function volumeCylinder($radius, $height) {
      return pi() * pow($radius, 2) * $height;
    }

    function volumeCone($radius, $height) {
      return (1/3) * pi() * pow($radius, 2) * $height;
    }

    function volumeSphere($radius) {
      return (4/3) * pi() * pow($radius, 3);
    }

    $side = 4;
```

```

$length = 5;
$width = 3;
$height = 7;
$radius = 6;

echo "<tr>
    <td>Cube</td>
    <td>V = s3</td>
    <td>s = $side</td>
    <td>" . number_format(volumeCube($side), 2) . "</td>
</tr>";

echo "<tr>
    <td>Right Rectangular Prism</td>
    <td>V = l × w × h</td>
    <td>l = $length, w = $width, h = $height</td>
    <td>" . number_format(volumeRectangularPrism($length, $width,
$height), 2) . "</td>
</tr>";

echo "<tr>
    <td>Cylinder</td>
    <td>V = π × r2 × h</td>
    <td>r = $radius, h = $height</td>
    <td>" . number_format(volumeCylinder($radius, $height), 2) .
"</td>
</tr>";

echo "<tr>
    <td>Cone</td>
    <td>V = (1/3) × π × r2 × h</td>
    <td>r = $radius, h = $height</td>
    <td>" . number_format(volumeCone($radius, $height), 2) . "</td>
</tr>";

echo "<tr>
    <td>Sphere</td>
    <td>V = (4/3) × π × r3</td>
    <td>r = $radius</td>
    <td>" . number_format(volumeSphere($radius), 2) . "</td>
</tr>";
?>

```

```
</table>
</body>
</html>

<style>
  body {
    font-family: Arial, sans-serif;
    background: #f0f8ff;
    padding: 20px;
  }

  h1 {
    text-align: center;
  }

  table {
    width: 60%;
    margin: auto;
    border-collapse: collapse;
    background: #fff;
  }

  th, td {
    padding: 10px;
    border: 1px solid #333;
    text-align: center;
  }

  th {
    background-color: #add8e6;
  }
</style>
```

Output:

Volume of Shapes

Shape	Formula	Values	Volume
Cube	$V = s^3$	$s = 4$	64.00
Right Rectangular Prism	$V = l \times w \times h$	$l = 5, w = 3, h = 7$	105.00
Cylinder	$V = \pi \times r^2 \times h$	$r = 6, h = 7$	791.68
Cone	$V = (1/3) \times \pi \times r^2 \times h$	$r = 6, h = 7$	263.89
Sphere	$V = (4/3) \times \pi \times r^3$	$r = 6$	904.78

Code 3:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Technical 2 Act 3</title>
</head>
<body>
<header>
  <h1>My Resume</h1>
  <nav>
    <a href="Resume.php?page=career">Career Objective</a>
    <a href="Resume.php?page=personal">Personal Info</a>
    <a href="Resume.php?page=education">Education</a>
    <a href="Resume.php?page=skills">Skills</a>
    <a href="Resume.php?page=affiliation">Affiliation</a>
    <a href="Resume.php?page=work">Work Experience</a>
  </nav>
</header>

<style>
html, body {
  font-family: Arial;
  height: 100%;
  margin: 0;
  background: white;
}

header {
```

```

        background-image: url('images/bg.jpg');
        color: white; padding:
        20px; text-align:
        center;
    }

    nav a {
        margin: 0 15px;
        text-decoration: none;
        color: white;
        font-weight: bold;
    }

    section {
        padding: 20px;
        background: white;
        margin: 20px;
        height: 69vh;
    }

    .wrapper {
        flex: 1;
    }

    footer {
        text-align: center;
        background-image: url('images/bg.jpg');
        color: white;
        padding: 10px;
    }
</style>

```

```

<div class="wrapper">
    <footer>
        <p>&copy; 2025 My Resume. All Rights Reserved.</p>
    </footer>
</div>
</body>
</html>

```

```

<?php
require('requires/header.php');

```



```

$page = $_GET['page'] ?? 'personal';

switch ($page) {
    case 'career':
        include('includes/career.php');
        break;
    case 'education':
        include('includes/education.php');
        break;
    case 'skills':
        include('includes/skills.php');
        break;
    case 'affiliation':
        include('includes/affiliation.php');
        break;
    case 'work':
        include('includes/work.php');
        break;
    default:
        include('includes/personal.php');
        break;
}

require('requires/footer.php');
?>

```

```

<section style="font-size: 24px; line-height: 1.6;">
  <h2 style="font-size: 28px; color: #004080;">Affiliations</h2>
  <ul>
    <li>Student, Far Eastern University - Institute of Technology</li>
  </ul>
</section>

```

```

<section style="font-size: 24px; line-height: 1.6;">
  <h2 style="font-size: 28px; color: #004080;">Career Objective</h2>
  <p> To obtain a challenging position in a reputable organization where I can
  apply and develop my knowledge in Information Technology, particularly in
  cybersecurity. I aim to contribute to the growth of the company while continuing
  to learn and improve my skills through hands-on experience, teamwork, and
  continuous learning.</p>
</section>

```

```

<section style="font-size: 24px; line-height: 1.6;">
  <h2 style="font-size: 28px; color: #004080;">Educational Attainment</h2>
  <ul>
    <li><strong>College:</strong> Far Eastern University - Institute of
    Technology - BS Information Technology Specialization in Cybersecurity</li>
    <li><strong>Senior High:</strong> Romblon State University - Laboratory
    Science High School</li>
    <li><strong>Junior High:</strong> Romblon State University - Laboratory
    Science High School</li>
  </ul>
</section>

```

```

<section>
  <div style="display: flex; font-size: 28px; align-items: center; gap: 20px;">
    
    <div>
      <p><strong>Name:</strong> Franzese Denn B. Selosa</p>
      <p><strong>Email:</strong> franzese.denn.selosa@gmail.com</p>
      <p><strong>Phone:</strong> 09602651553</p>
      <p><strong>Address:</strong> Sampaloc, Manila, Metro Manila</p>
    </div>
  </div>
</section>

```

```

<section style="font-size: 24px; line-height: 1.6;">
  <h2 style="font-size: 28px; color: #004080;">Skills</h2>
  <ul>
    <li>HTML and CSS</li>
    <li>PHP</li>
    <li>Python</li>
    <li>Cybersecurity Basics</li>
    <li>Teamwork and Communication</li>
  </ul>
</section>

```

```

<section style="font-size: 24px; line-height: 1.6;">
  <h2 style="font-size: 28px; color: #004080;">Work Experience</h2>
  <ul>
    <li>None</li>
  </ul>
</section>

```

Output:



VII. QUESTION AND ANSWER

1. What is Array?
An array is a special type of variable in PHP that can store many values in one place. Instead of creating a separate variable for each item, you can group them together in an array.
2. What is function in PHP?
A function in PHP is a block of code that performs a specific task.
3. Differentiate the one-dimensional array to multi-dimensional array.
A one-dimensional array is like a simple list that holds values in a single row, such as a list of names or numbers. A multi-dimensional array is more like a table or grid, where each element can also be an array itself.
4. Describe the difference between Global Variable to Local Variable.
A global variable is a variable that can be used anywhere in the code, even inside functions if we declare it as global. A local variable, on the other hand, only works inside the function or block where it was created. Outside that block, it doesn't exist.
5. What is the difference between function without parameter and a function with parameter?
A function without a parameter does not need any extra information to run, it just does the same thing every time. A function with a parameter needs input or data to work with, so we can change how it behaves based on the values we give it.
6. What are the different common predefined functions? Describe each and give example.
strlen() – This counts the number of characters in a string. Example: strlen("Hello") returns 5.
strtoupper() – Changes all letters in a string to uppercase. Example: strtoupper("hello") becomes "HELLO".
count() – Counts how many items are in an array. Example: count(\$fruits) returns the total number of items.

7. Is it useful to use an include or require function on a webpage? Why?
Yes, using include or require is very useful because it helps us organize our code by putting reusable parts, like headers, footers, or menus, in separate files.
8. What is the importance of a function?
Functions are important because they break our program into smaller, manageable parts. They make our code easier to read, fix, and reuse. Instead of repeating the same code many times, we can just call the function whenever we need it.
9. What is the importance of a string function?
String functions are important because they help us handle and change text in our program. We can use them to count letters, change case, find words, or join text.

VIII. REFERENCES

1. <https://www.w3schools.com/css/>
2. <https://www.w3schools.com/html/>
3. https://www.w3schools.com/php/php_variables.asp
4. https://www.w3schools.com/php/php_arrays.asp
5. https://www.w3schools.com/php/php_arrays_indexed.asp
6. https://www.w3schools.com/php/php_arrays_associative.asp
7. https://www.w3schools.com/php/php_arrays_multidimensional.asp
8. https://www.w3schools.com/php/php_arrays_sort.asp
9. https://www.w3schools.com/php/php_functions.asp
10. https://www.w3schools.com/php/php_constants.asp
11. https://www.w3schools.com/php/php_includes.asp
12. https://www.w3schools.com/php/php_string.asp

Note: The following rubrics/metrics will be used to grade students' output.

Program (100 pts.)	(Excellent)	(Good)	(Fair)	(Poor)
Program execution (20pts)	Program executes correctly with no syntax or runtime errors (18-20pts)	Program executes with less than 3 errors (15-17pts)	Program executes with more than 3 errors (12-14pts)	Program does not execute (10-11pts)
Correct output (20pts)	Program displays correct output with no errors (18-20pts)	Output has minor errors (15-17pts)	Output has multiple errors (12-14pts)	Output is incorrect (10-11pts)
Design of output (10pts)	Program displays more than expected (10pts)	Program displays minimally expected output (8-9pts)	Program does not display the required output (6-7pts)	Output is poorly designed (5pts)
Design of logic (20pts)	Program is logically well designed (18-20pts)	Program has slight logic errors that do no significantly affect the results (15-17pts)	Program has significant logic errors (3-5pts)	Program is incorrect (10-11pts)
Standards (20pts)	Program code is stylistically well designed (18-20pts)	Few inappropriate design choices (i.e. poor variable names, improper indentation) (15-17pts)	Several inappropriate design choices (i.e. poor variable names, improper indentation) (12-14pts)	Program is poorly written (10-11pts)
Delivery (10pts)	The program was delivered on time. (10pts)	The program was delivered a day after the deadline. (8-9pts)	The program was delivered two days after the deadline. (6-7pts)	The program was delivered more than two days after the deadline. (5pts)